ILMEE DE SILVA

Full-Stack Software Engineer · Backend, Microservices, AI & Cloud · Python | Java | FastAPI | AWS | Azure

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PROFESSIONAL SUMMARY

Full-stack software engineer with **3+ years** of production experience, focused on the backend: Python, FastAPI, Java, SQL, microservices and AWS/Azure cloud integrations. Holds a **First Class BSc (Hons) in Computing**; final-year research built **MESA**, a cross-service static analysis tool that found every labelled dependency in three real microservice systems with no false positives, where two frontier LLMs recalled at most 25%. Builds and publishes AI tooling, including **patchpilot**, an open-source LLM agent on PyPI that fixes the code a dependency upgrade breaks and opens the pull request. Experienced across the full delivery lifecycle: development, QA, release management and UAT.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Swift, XML/XSD

Backend & APIs: FastAPI, Spring Boot, Node.js, Express, REST API design, JWT authentication, OpenAPI/Swagger, Pydantic, Prisma, SQLAlchemy, Postman

Microservices & Architecture: microservice design, service boundaries, anti-pattern detection, static analysis (AST), graph modelling (NetworkX, Neo4j), architecture fitness functions and CI gates

AI & ML: LLM agents & tool use (Claude API), prompt chains with structured output, RAG, Azure AI Search (semantic & hybrid), intelligent document processing, PyTorch, transfer learning

Cloud & DevOps: AWS (Lambda, API Gateway, S3, CloudWatch), Azure (AI Search, Blob Storage), Docker, Docker Compose, GitHub Actions, CI/CD, Azure DevOps, Vercel

Data: SQL Server stored procedures, PostgreSQL, Neo4j, Apache Airflow, ETL pipelines, data modelling, validation & normalisation

Frontend & Mobile: React, Next.js, Tailwind CSS, SwiftUI, Core Data

Practices: Agile/Scrum, SDLC, Design Science Research, QA & regression testing, UAT, release management, technical documentation

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | Novigi Pty Ltd, Australia & Sri Lanka *Jun 2023 – Present*

AI-Tax DataSmart: AI Document Extraction Pipeline

Python, FastAPI, Azure AI Search, Azure Blob Storage, LLMs, SQL Server

- Architected and built Python/FastAPI services that extract structured tax data from unstructured, multi-page superannuation fund statements with inconsistent layouts.
- Part of the team that turned a month-long manual tax-data review into a matter of days with an automated, AI-driven extraction pipeline, reducing processing time and human error in a compliance-critical workflow.
- Designed Azure AI Search indexes (schemas, searchable fields, scoring profiles) over Azure Blob Storage for semantic and hybrid retrieval of PDF content, and introduced a reusable partitioning strategy to work around index limits.
- Combined LLM prompt chains and structured output parsing with algorithmic search to reliably locate specific financial values, including small numeric values the previous approach missed.
- Built and consumed REST APIs linking the extraction engine, processing pipeline and downstream reporting, covering contract design, authentication and retry/error handling.
- Wrote SQL Server stored procedures to automate multi-stage extraction, validation and classification, enforcing business rules and flagging exceptions for compliance review.
- Worked directly with domain experts to translate Australian superannuation tax-reporting requirements into data models and extraction logic.

Apex Corro: SmartCommunications Implementation

SmartCommunications, XML/XSD, data mapping

- Built and configured automated client-correspondence letter templates in SmartCommunications for Australian Acurity Funds.
- Designed XML/XSD data models to keep field mapping between SmartCommunications and Acurity accurate across regulated fund correspondence.
- Translated business requirements into template designs and data-mapping specifications with stakeholders.

Internal Platforms & Integrations

AWS Lambda, Apache Airflow, REST APIs, Postman, Azure DevOps

- Built and maintained AWS Lambda-based integrations connecting Zoho, ProjectWorks, Employment Hero, Xero and BambooHR.
- Built Apache Airflow pipelines that sync data across platforms and databases, improving the accuracy and efficiency of cross-system reporting.
- Served as release manager: coordinated production releases, prepared release notes, ran UAT demos for stakeholders and managed support handovers.
- Owned regression testing and QA for internal projects, and wrote technical documentation that reduced onboarding time for new team members.

Prior Experience | before moving into software engineering

Personal Assistant to CEO | Ndurance (Pvt) Ltd – Yeti Isotonic, Colombo — *Jun 2022 – Jun 2023*
Secretary to Managing Director | Micronet Global Services (Pvt) Ltd, Colombo — *Sep 2021 – Jun 2022*
Training & Development Assistant | Sri Lanka Badminton Association, Colombo — *Jan 2021 – Aug 2021*
Executive Sports / Executive Projects | McLarens Holdings Limited, Colombo — *Jun 2015 – Dec 2020*

FINAL-YEAR RESEARCH

MESA | Cross-service static analysis for microservice architectural anti-patterns — *2025 – 2026*

Python, FastAPI, NetworkX, Neo4j, PostgreSQL, React, TypeScript, Docker · github.com/ilmeedesilva/MESA

Paper in preparation for publication (K. I. W. De Silva, M. W. Wadumulla)

- Designed and built, through Design Science Research, a tool that turns multiple microservice repositories into a service interaction graph without executing code, and detects five anti-patterns (cyclic dependencies, shared databases, chatty services, hard-coded dependencies, God services) with evidence and limitations on every finding.
- Reproduced all 76 seeded findings across 8 benchmark systems; on three real public systems (Online Boutique, a Spring Boot sample, Sock Shop) found all 30 labelled dependencies with 0 false positives after two benchmark-verified fixes, in under 0.6 s per corpus.
- Compared against Claude Sonnet 5 and Claude Opus 5 over five runs per system: MESA was fully deterministic, while both LLMs recalled at most 25% on the real systems and varied between runs. Added health scores, governance policies and a CI fitness gate.

SELECTED PROJECTS

patchpilot | Open-source AI agent for dependency upgrades — *2026*

Python, Claude API (tool use), Docker, Git worktrees, GitHub Actions, PyPI · github.com/ilmeedesilva/patchpilot

- Built an LLM agent that upgrades a Python dependency inside a locked-down Docker sandbox, reads failing tests and changelogs, fixes the broken code and opens a pull request describing every breaking change.
- Hand-wrote the tool-use agent loop; patchpilot re-runs the tests itself before committing and opens failing upgrades as draft PRs. Published on PyPI with CI; runs weekly as a GitHub Action and has opened PRs unattended (e.g. marshmallow 3 → 4).

DR-Triage | Explainable diabetic retinopathy screening — *2026*

PyTorch, EfficientNetV2-S, OpenCV, Grad-CAM++, RAG · github.com/ilmeedesilva/Diabetic-Retinopathy-Stage-Detection

- Trained an order-aware (CORN ordinal) CNN with transfer learning on 3,662 clinician-graded fundus images: test QWK 0.854 and referable-DR AUC 0.979, with calibrated confidence and MC-Dropout uncertainty.
- Added an image-quality gate, Grad-CAM++ explanations and a retrieval-grounded chatbot that answers with cited screening-guideline snippets.

EDUCATION

BSc (Hons) in Computing – First Class Honours | Coventry University — *2025 – 2026*
Higher National Diploma in Software Engineering | NIBM, Colombo · GPA 3.94 — *2024*
Diploma in Software Engineering | NIBM, Colombo · GPA 4.0, Batch Top, Gold Medalist — *2023*
Visakha Vidyalaya | Colombo 05 — *2010 – 2015*
Yasodara Balika Vidyalaya | Colombo 08 — *2002 – 2010*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Cloud:** AZ-900 Microsoft Azure Fundamentals (in progress); Microsoft Azure Data Fundamentals (DP-900) Exam Prep Specialization, Microsoft (2024); Azure SQL, Azure Cosmos DB, Core Data Concepts and Modern Data Warehouse Analytics in Azure, Microsoft (2024)
- **Data & Python:** Python for Data Science and AI, Coursera (2024); Python Basics, University of Michigan (2025); Introduction to SQL Window Functions, Coursera (2024); Machine Learning with Apache Spark, IBM (2024)
- **Web & AI:** Learn Next.js, Scrimba (2026); Foundations of Prompt Engineering for Generative AI

TECHNICAL WRITING

Articles on dev.to/ilmeedesilva and ilmeedesilva.vercel.app: building an AI agent that fixes dependency upgrades, semantic search with Azure AI Search, solving index limits with partitioning, and combining AI with algorithmic search for financial values.

ACHIEVEMENTS

- **Junior and senior national badminton player, representing Sri Lanka internationally:** World Junior U-19 (Malaysia, 2014), Asian Junior U-19 (Malaysia, 2013), ANA Asia Youth U-17 & U-15 (Japan, 2010–2011); Women's U-25 Champion, 2nd Student Olympic International Games (2015–16).
- **National:** National Open U-17 Singles Champion (2013); multiple All-Island age-group champion (2006–2014); Sri Lanka Schools Badminton Colours, Merit Awards (2009–2014).